

N=29 Murty-Simon candidate: verification companion

Reviewer-v3 evidence, replay map, and audit boundaries

Paul Lenz

11 September 2026

11 September 2026. Research directed by Paul Lenz; mathematical development, implementation and internal checking by ChatGPT/Geeps.

Status: candidate mathematics. Independent mathematical review, independent computational reproduction and novelty assessment remain OPEN.

1. Purpose

This companion maps the evidence layer for the current reviewer-v3 candidate proof that every simple diameter-two edge-critical graph on 29 vertices satisfies

$e(G) \leq 210$,
with equality exactly $K(14,15)$.

The current proof is deliberately narrower than the historical route. Reviewers do not need to validate every exploratory program preserved in the repository.

2. Current proof architecture

Scope	Current method	Proof-critical computation?
$\Delta \leq 14$	degree sum	no
$\Delta = 15$	witness-deficit hand proof and equality	no
$\Delta = 16, m = 210, 211$	minimal trusted kernel + exact Farkas	yes
$\Delta = 16, m = 212$	trusted-kernel extension + exact Farkas	yes
$\Delta = 16, m = 213, 214$	exact threshold/source-count/Hall arithmetic	small finite check
$\Delta = 16, m = 215$	hand threshold-capacity contradiction	no
$\Delta = 16, m \geq 216$	pointwise $5/2$ charging cap	no
$\Delta = 17$	pointwise charging hand bound	no
$\Delta = 18, \dots, 27$	residual h-index hand bound	no

Scope	Current method	Proof-critical computation?
Delta = 28	universal-vertex/star observation	no

The old generic scan over $m=216..232$ is now corroborative evidence only, not a logical dependency.

3. Main graph-to-model trust boundary

The self-contained bridge is

`project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.md`.

Its proof-critical universal statements are:

1. exactly one selected cross quasi-edge is designated for each missing **unordered** B-pair;
2. the forced cross-edges used by the injections cannot secretly be selected when their endpoints jointly miss an A-vertex;
3. residual activity $\rho_u \geq 1$ holds for every B-source when $t > 0$;
4. selected-source demand satisfies $s_i \leq \rho_u$;
5. the charging inequality is necessary;
6. the threshold-capacity inequality is necessary;
7. the isolated-C exclusion is valid;
8. every actual graph satisfying the bridge maps into the corrected late linear relaxation.

A counterexample to any universal bridge lemma overrides all downstream computations that depend on it.

4. Minimal trusted kernel at 210 and 211

The preferred dense computation is recorded in

`project/reviews/n29/2026-09-08-redteam-restart-v1/MINIMAL_KERNEL_REPORT.json`.

Edge count	t	Retained demands	Residual rows	Exact late rejections	Final survivors
211	3	72	126	126	0
210	2	367	1,467	1,467	0

The proof-critical implementation is intentionally small:

- `minimal_prepare.py` - exact demand preparation and early exact cuts;
- `minimal_rows.cpp` - complete sorted residual-row enumeration with necessary Hall/refinement cuts;
- `run_minimal_kernel.py` - terminal certificate runner;
- `independent_threshold_model_v2.py` - corrected cumulative-threshold/source-q-flow model;
- the exact certificate machinery in `independent_threshold_model.py`.

The route does **not** depend on the old projected screen, joint propagator, shared LP, typed LP, endpoint LP, or older pair-capacity support formula.

5. Historical normalization bug

The first additional cumulative-threshold builder contained a real normalization error: label-group multiplicity was counted twice on the label side of the selected-incidence equations.

Accordingly:

- v1 cumulative-threshold certificates are invalid as proof evidence;
- the flawed v1 source remains preserved for audit history;
- `independent_threshold_model_v2.py` reconstructs the graph-derived builder with the corrected per-label normalization;
- the minimal trusted kernel uses v2;
- every terminal exclusion is rechecked by exact integer arithmetic.

The corrected selected-incidence normalization is schematically

$$\sum_k n_k Z = E[x] = \sum_h T_h,$$

with no extra multiplication by the label-group size.

6. Exact Farkas semantics

Floating-point LP output is used only to **propose** a ray. Solver status alone is never an exclusion.

For nonnegative variables, the checker accepts only an exact integer combination in which:

- multipliers on $\leq$ inequalities are nonnegative;
- equality multipliers may be signed;
- the combined coefficient of every variable is nonnegative;
- the combined right-hand side is strictly negative.

This produces an inequality with nonnegative left side and negative right side, an exact contradiction. The final check uses integer arithmetic.

7. Upper-range finite checks still used by reviewer-v3

For $\Delta=16$, $t=m-208$.

At $m=212$ ($t=4$):

```
2032 charging-feasible profiles
1706 threshold rejects
 258 source-count rejects
   65 exact early Hall-dual rejects
    3 open demand profiles
19630 residual numerical states
    2 residual-row survivors
    2 exact late Farkas rejections
    0 final survivors
```

At $m=213$ ($t=5$), all 586 profiles are excluded by 576 threshold cuts, 7 source-count cuts and 3 exact Hall duals.

At $m=214$ ($t=6$), all 79 charging-feasible profiles are threshold-rejected.

No finite verifier is logically required at $m=215$ or above.

8. Blind hostile review and independent reruns

A separate ChatGPT instance without the project background performed a hostile review supplied by the user. It reported no fatal flaw after independently attacking the graph-to-model bridge, residual activity, threshold capacity, isolated-C, corrected LP normalization, exact Farkas semantics, the witness equality argument, the $\Delta=17$ bound, the residual h-index calculation and the cited dominating-edge theorem.

The preserved follow-up is

`project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/FOLLOWUP.md`.

Charging-domain census

The blind reviewer independently obtained the complete nondecreasing demand counts

```
t=2: 9251
t=3: 4867
t=4: 2032
t=5:  586
t=6:   79
t=7:    1
t=8:    0
```

and the project regenerated exactly the same sequence again. The rerunnable check is

`project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/independent_charging_census.py`.

Graph-atlas bridge regression

The blind reviewer reported an exhaustive NetworkX graph-atlas check through order 7. The project independently reproduced:

```
21 unlabeled diameter-two-edge-critical isomorphism types
50 maximum-degree rooted cases
58 admissible selection configurations
0 bridge failures
```

The regression checks the exact ledger, demand implication, selected-edge injections, supplement forcing, source capacities, endpoint load and selected-source demand. All small examples have t in $\{-3, -2, -1, 0\}$, so this does **not** test positive-surplus residual activity, charging or threshold capacity.

The rerunnable script is

`project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/graph_atlas_bridge_regression.py`.

9. Analytic upper-range hardening

For integer $0 \leq s \leq 11$, reviewer-v3 uses

$$s(13-2s)/(12-s) \leq 5/2,$$

with equality only at $s=4$. Therefore the twelve-label charging sum is at most 30, while the bridge requires at least $16+2t$. Hence $t \leq 7$, or $m \leq 215$.

At $m=215$, equality forces all twelve demands to equal four. The charging lower bound and $S \geq r+2t$ force $r=34$; residual activity gives $z_4 \leq 6$; threshold capacity would require

$$96 \leq z_4^2 - z_4 + 20 \leq 50,$$

a contradiction.

Thus the historical $m=216 \dots 232$ scan and the former $m=215$ finite check are no longer proof-critical.

The cross-order propagation is preserved in

project/research/fan-free-fixed-orders/2026-09-11-pointwise-analytic-caps-v1/POINTWISE_CAPS.md

10. Remaining correlated-error risk

The largest remaining correlated-error risk is the **actual-graph-to-averaged-variable embedding of the corrected late LP**. The current model has survived hostile dimensional and normalization review, but an independent mathematician/programmer should reconstruct this mapping from scratch.

The next most valuable independent checks are residual activity, threshold capacity, isolated-C, and an independently authored exact verifier.

11. Recommended reviewer order

A hostile independent audit should proceed in this order:

1. reconstruct the complement/quasi-edge construction;
2. attack selected-edge and forced-residual injections for collisions;
3. reconstruct residual activity;
4. reconstruct $s_i \leq \rho_u$ and charging;
5. reconstruct threshold capacity;
6. reconstruct isolated-C;
7. map an actual graph into every grouped variable of the corrected late model;
8. independently implement the late linear relaxation;
9. verify a sample and then all exact Farkas certificates;
10. regenerate the finite frontiers from independently written code.

12. Canonical files

Current proof sources:

- project/reviews/n29/2026-09-11-reviewer-v3/PROOF.md
- project/reviews/n29/2026-09-11-reviewer-v3/GRAPH_TO_MODEL_BRIDGE.md

Evidence and audits:

- `project/reviews/n29/2026-09-08-redteam-restart-v1/MINIMAL_KERNEL_REPORT.json`
- `project/reviews/n29/2026-09-08-redteam-restart-v1/PUBLIC_RELEASE_AUDIT.md`
- `project/reviews/n29/2026-09-11-blind-external-ai-redteam-v1/FOLLOWUP.md`
- `project/reviews/cross-cutting/2026-09-11-n29-cross-order-reaudit-v1/REPORT.md`
- `project/research/fan-free-fixed-orders/2026-09-11-pointwise-analytic-caps-v1/POINTWISE_CA`

Historical reviewer-v2 PDFs remain preserved for provenance but are superseded by reviewer-v3.